SDC

IIC S.P.I.T.
Institution's Innovation Council

SARDAR PATEL INSTITUTE OF TECHNOLOGY

DataForge

A 6 Hour ML Hackathon

PROBLEM STATEMENTS

PS 1

Computer Vision

ART-IQ

From Van Gogh's swirls to Picasso's abstracts, each artist and art movement has a distinct visual fingerprint. Art historians can often identify an artwork's style at a glance—but can machines do the same?

Your challenge is to build ArtIQ, a model that can analyze an artwork (sketch, drawing, painting, animation, comic/manga) and classify its art style based solely on visual cues.

Core Functionalities:

1. Art Style Detection Model
   - Build a computer vision model capable of recognizing and classifying different artistic styles from input images.
   - Styles can range across digital aesthetics (e.g., Ghibli-style, pixel art), traditional styles (e.g., Van Gogh, Baroque), or material-driven styles (e.g., Yarn art, Collage).
   - The model should be robust to color, stroke, texture, and compositional patterns.
2. Flexible Classification Architecture
   - Use a CNN-based architecture or transfer learning (e.g., ResNet, VGG, EfficientNet) to learn

visual features that differentiate artistic styles.

   - Design your model to scale with new styles or categories if added later.
3. Preprocessing and Input Pipeline
   - Create an efficient pipeline to process and normalize input images:
   - Resize, crop, or enhance artwork to ensure consistent model input
   - Apply relevant data augmentation techniques to improve generalization
4. Evaluation & Feedback
   - Display the predicted art style in a clear, interactive format (e.g., web UI or console).

PS 2
Computer Vision
ATTENTION SPAN!

Long videos are everywhere — lectures, podcasts, vlogs, surveillance footage, even sports. Watching or analyzing hours of video is exhausting. What if you could build an AI system that automatically summarizes a video into its key moments, not just using audio or text, but purely through visual activity?

This system should process a long video and generate a short visual storyboard or highlight reel by identifying and extracting the most visually significant segments — such as:

- High motion activity
- Sudden scene changes
- Entry or exit of people
- Unique object appearances

Core Functionalities:

- Accept a video input and break it into frames or clips.
- Analyze visual motion patterns, scene transitions, and object/person detection.
- Score and rank segments based on visual significance.

PS 3

Natural Language Processing (NLP)

GREAT! I LOVE IT!

In the wild world of social media, sarcasm is the ultimate curveball for machines. A tweet that says “Great, another Monday 🙃” might actually mean the opposite of what it says.
Your task is to build an Explainable AI system that can analyze short-form social media posts (e.g., tweets) and determine whether the post is genuine or sarcastic.
This involves understanding nuanced human emotions, cultural tone, irony, and the context that flips a sentence's meaning. It's
not just what’s written — it’s how it's written.

Core Functionalities:

1. Sarcasm Identification Engine
   - Develop a system that accepts a short social media post as input and predicts whether it is sarcastic or genuine.
   - Output should Include:-
     - A binary label: Sarcastic or Genuine
     - A confidence score indicatin prediction certainty
     - An explanation component that highlights which words, emojis, or linguistic features contributed most to the classification
2. Text Understanding Pipeline
   - Design a pipeline that captures the underlying tone, hidden meanings, and linguistic cues present in informal language.
   - Handle diverse elements like emojis, hashtags, elongated words (soooo), and punctuation-based sarcasm (great!!!).

3. Language Intelligence Module

- Your model should move beyond just surface-level word matching. Aim to understand implicit intent, contrast in sentiment, or contextual irony embedded in user-generated content.
- Employ meaningful text representations that can differentiate between “love it” and “LOVE it 🙄”.

4. Explainability Interface

Provide users (e.g., researchers, moderators, or users) with:

- A visual or textual explanation of the model’s reasoning (e.g., highlighted tokens or phrases)
- Options to adjust threshold sensitivity or view top sarcasm indicators
- Ability to compare sarcastic and genuine examples side by side

PS 4

Natural Language Processing (NLP)

RESEARCH FORGE

Researchers often struggle to find the right journal to submit their work to — whether due to mismatched scope, low acceptance chances, or simply being unaware of niche publications.
Your task is to build an AI-powered recommendation system that, given the title and abstract of a research paper, can suggest the top 3 most relevant academic journals where the paper is most likely to be a good fit.
Think of it as a "Spotify for researchers" — matching their content with the best-suited outlets based on topics, tone, technical depth, and scope.

Core Functionalities:

Research Content Understanding Engine

- Accept a research paper's title and abstract (or a small summary) as input.
- Extract meaningful embeddings, key topics, and semantic context from the content.

2. Journal Matching Model

- Compare the research paper against a curated dataset of journal metadata.
- Use similarity measures, embedding-based matching, or classification models to suggest
  the top 3 journals.
- Ensure recommendations are ranked based on relevance and scope-fit.

3. User Interface (Bonus)

- Simple frontend where users paste their title + abstract and receive journal suggestions.
- Bonus: Let users filter results by domain or impact factor range.

Sample data API : https://docs.openalex.org/

PS 5

Generative AI (GenAI)

HUNTER X HUNTER

Generally budding or aspiring authors have doubts or certain complications reagrding their plot/ storyline. An AI assistant that could help in making coherent and engrossing storylines and also help authors fix problems/ blocks in their plots could help. Also, it could act as a storywriter by writing a chapter or a piece of a chapter. The model should be able to interpret the target audience age of the novel/story/novella, generate the chapters accordingly using the audience-tailored vocabulary, should not generate gibberish and must be high quality descriptive storywriting by adhering to modern and loved writing practices by referring to some of the bestsellers across all genres.

Core Functionalities:

1. Audience-Aware Chapter Generation
   - Accept user inputs like story genre, age group of readers, plot outline, and character details.
   - Use LLMs to generate high-quality story content with vocabulary and tone appropriate for the specified audience (e.g., children, young adults, adults).
2. Narrative Coherence & Plot Block Assistance
   - Detect plot holes, inconsistencies, or pacing issues. Suggest changes and/or review current plots from authors.
   - Suggest alternate plot directions, character arcs, or climaxes.
   - Bonus: Use embeddings or memory tools to track long-term plot context.
3. Creative Expansion & Branching
   - Offer multi-branching story path suggestions (e.g., "What if the villain betrays the hero?")
   - Allow writers to explore different outcomes in a “choose-your-own-adventure” format.

PS 6

Generative AI (GenAI)

MEMEMIND

Memes are the modern-day language of the internet—used for humor, satire, and social commentary. But what if you could automatically generate memes that are not only hilarious, but topically relevant, context-aware, and visually coherent?

Your challenge is to build MemeMind, an AI-powered meme generator that takes in a topic or real-world event and produces a meme image with appropriate layout, caption, and emotion—using computer vision and GenAI techniques.

Core Functionalities:

1. Topic-Aware Meme Generator
   - Accept input: a news headline, recent event, trending topic, or plain text idea.
   - Interpret the emotion/sentiment, cultural context, and tone of the topic.
2. Image + Caption Generation
   - Auto-generate a meme image:
     - Either from a pre-defined meme template bank
     - OR using text-to-image generation models (e.g., Stable Diffusion / DALLE)
   - Create a caption that aligns with the meme format and topic, using humor or satire.
3. Vision + Layout Alignment
   - Accurately place text on the meme image (top/bottom or in comic-style bubbles).
   - Ensure font, spacing, and color are meme-appropriate and readable.

PS 7

Open Innovation

EMERGENCY REPORTER

Emergencies can strike at any time and place, and effective resource allocation is crucial for minimizing response times and saving lives. With access to historical emergency call data (anonymized) and geospatial information, your task is to predict high-demand areas for emergency services and optimize resource allocation to ensure quicker response times.
The goal is to develop a system that leverages historical emergency data, including call types, times, locations, and more, to forecast future demand for emergency resources (e.g., ambulances, fire trucks). The model should be able to predict areas with high emergency call frequency and recommend optimal deployment strategies based on real-time data.

Core Functionalities:

1. Historical Data Analysis Engine
   - Use historical emergency data (e.g., time of day, date, call types, etc.) to identify patterns of demand for emergency services.
   - Handling long-term trends vs. short-term fluctuations: Emergency services can have long-term trends (e.g., rising healthcare demand), but short-term spikes in calls might happen due to specific incidents.
   - Anomaly Detection: Detecting outliers where calls increase dramatically due to a special event or emergency.
2. Geospatial Demand Prediction
   - Use location data to understand which areas experience the highest volume of emergency calls.
   - Combine time of day, seasonality, and historical trends to predict emergency hotspots.

PS 8

Open Innovation

SKILLCOIN

The goal of this challenge is to design a decentralized skill-sharing platform where users exchange services through a time-based credit system, SkillCoins. The platform should be intelligent, matching users who offer and request specific skills. Reinforcement learning (RL) should be used for the matching process, ensuring that users are matched optimally based on their preferences, availability, and trust scores.
The system should also incorporate trust scoring and fraud prevention features to ensure the safety and reliability of transactions.

Core Functionalities:

1. Skill Exchange System:
   - Users offer services (skills) they are proficient in and request services (skills) they need.
   - Users are rewarded with SkillCoins for offering their services, which they can then spend on other users' services.
   - Each user’s available hours for offering services should be tracked to prevent misuse of resources.
2. Trust Scoring:
   - Each user has a trust score based on their previous interactions. Positive feedback and successful skill exchanges improve the trust score.
   - Trust scoring can be dynamic and should be updated in real-time after each interaction.
   - The system should have fraud prevention mechanisms, such as detecting discrepancies between claimed skills and actual service delivery (possibly through feedback loops and verification).